Intrinsic Hyperspectral Image Decomposition With DSM Cues

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***Abstract*— Intrinsic hyperspectral image decomposition (IHID) aims to recover physical scene properties such as reflectance and illumination from a given hyperspectral image (HSI), which directly respects the physical imaging process and can benefit many HSI processing tasks. It is a severely ill-posed problem and is challenging to solve using HSI alone. Additional geometric information provided by digital surface models (DSMs) can otherwise help immensely. While intrinsic image decomposition for RGB images and RGB-D images has been studied extensively during the past few decades and has seen significant progress, studies of the problem for other types of data, such as HSIs and DSMs, are still needed. It is much more challenging to handle an HSI with hundreds of channels than an RGB image with only three channels. Moreover, compared with RGB-D data, HSIs and DSM data usually have much lower spatial resolutions and more complicated land covers, making it difficult to extend the RGB-D intrinsic image method directly. In this article, we present a novel IHID framework for HSIs with DSM cues. Utilizing spherical-harmonic illumination, we first propose a convenient HSI rendering model with DSM, which describes the interplay of material reflectance, geometric distribution, and environment illumination. Then, we introduce local and nonlocal priors on reflectance that ensure the local smooth and global consistency of recovered reflectance. Experiments on synthetic and real data demonstrate that the proposed method outperforms the state-of-the-art methods and is robust to illumination changes.**

***Index Terms*— Data fusion, digital surface model (DSM), hyperspectral images (HSIs), intrinsic hyperspectral image decomposition (IHID).**

I. Introduction

REMOTE sensing technology has made rapid advances in recent decades. Currently, the following vast range of instruments is available; many different types of data can be acquired by various imaging sensors mounted on various platforms. The surge in the diversity and amount of data has made data fusion a hot topic in the remote sensing field. In terms of inputs and applications, there are many types of data fusion working in different ways. Among the many fusion tasks, one that has drawn special interest is the combination of spectral and elevation information. Typically, hyperspectral images (HSIs) are fused with light detection and ranging (LiDAR) data that are highly complementary to each other. Many HSIs and LiDAR data fusion methods have been proposed and have boosted various applications. These methods mainly focus on fusion at the feature and decision levels as follows. Feature-level methods aim to design useful spectral and geometric features [1]–[4], whereas decision-level methods aim to design sophisticated classifiers that can make better use of these features [5]–[8]. Recently, HSI and LiDAR fusion using convolutional neural networks (CNNs) has become a hot topic [9], [10]. They ignore the fact that environmental elements, such as illumination and shadow, can significantly impact the observed spectrum and reduce the robustness of the extracted features. HSIs have the problem of high spectral dimensionality [11] and high spectral variability [12]. Hong *et al.* [13], [14] further proposed a multimodal deep learning fusion method that utilizes deep networks to unify diverse fusion architectures and achieves good results.

Although designing illumination-invariant features [15]–[17] can be a possible solution to this problem, a more straightforward and appealing idea is that we can try to model the imaging process utilizing the spectral and geometric distribution of the observed scene. Then, we can pursue the reverse problem to estimate intrinsic properties such as reflectance and environmental properties such as illumination from the image, which is also called the intrinsic image decomposition (IID) problem. The advantages of implying IID to combine spectral and elevation information are as follows: 1) it directly respects how an image is physically generated and 2) estimated scene properties, such as reflectance and illumination together with elevation, are intuitively essential to understand the real physical world.

The concept of IID was first proposed in the computer vision field and refers to the separation of the reflectance component and shading component from a single RGB image [18]–[21], which can be expressed as

$$I = s + R \tag{1}$$

where I is the log image, s denotes the log-shading component, and R denotes the log-reflectance component. R is only dependent on the material's intrinsic characteristic and is invariant to illumination conditions, while the shading component s measures the amount of light reflected on a certain point. This is a typically ill-posed problem; the key to acquiring reasonable decomposition results is to propose good constraints that are based on prior knowledge. For example, one extensively used assumption is that if two neighboring pixels have similar chromaticity, they also have similar reflectance.

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A more general form of the "intrinsic image" problem is described as the "shape, illumination, and reflectance" problem, which further decomposes the shading problem into shape and illumination. This can be done by depth prediction using a novel CNN [22]–[24] or by natural image statistics analysis [25], and the latter usually requires the inputs to be imaged with only single-segmented objects. For the more general case, i.e., recovering the shape, illumination, and reflectance from the whole image scene where objects may occlude or support one another and spatially varying illumination exists, additional sources, either image sequences illuminated under varying illumination [26]–[28], depth cues acquired by depth sensor [29]–[33], or incident light direction [34], are needed. Among the many types of additional sources, depth data, which provide geometric information complementary to color information, are the most useful and powerful.

While there are various effective IID models for RGB and RGB-D images, IID methods for other types of data, such as HSIs, are still needed. HSIs measure contiguous spectral bands from visible light to near-infrared light and finally form a hyperspectral cube containing much spectral information. Different from RGB images that have only three color channels, HSIs may contain hundreds of spectral bands. The challenge brought by the high spectral dimensionality is that the different bands of an HSI are highly correlated and redundant, making it difficult to generalize an IID model for RGB to HSI. Moreover, for a remote sensing HSI, the spatial resolution is relatively lower and will cause a spectral mixing phenomenon. IID from a single HSI has already been investigated by some previous works [35]–[37], which decomposes an HSI into a hyperspectral reflectance component and a shading component, which we call intrinsic hyperspectral image decomposition (IHID). These IHID methods have been demonstrated to be useful in HSI processing takes, such as classification [38] and resolution enhancement [39]. However, these methods still have the following drawbacks: 1) uniformly distributed illumination is assumed, ignoring the fact that spatially varying illumination may exist in a real scene; 2) shadow effects are ignored; 3) properties, such as illumination and elevation, are not recovered. Jin and Gu [40] and Gu *et al.* [41], [42] proposed a joint IID framework for HSIs and LiDAR point clouds, which can successfully infer scene properties such as illumination, reflectance, and surface normals from HSIs and LiDAR. However, it requires a high point density of LiDAR point clouds and utilizes LiDAR intensity as supplementary information. However, LiDAR data are not always available, especially for satellite HSIs. In this case, designing a model that can tackle other kinds of elevation data, such as digital surface model (DSM) data generated by satellite stereo-pair images, is still necessary. In this article, we present a novel IHID method with DSM cues.

In summary, our main contributions are given in the following.

A. Novel HSI Rendering Model

The existing HSI rendering model decomposes an HSI into a reflectance component and a shading component. Considering that the shading component is the interaction of illumination and surface normal, we propose a new HSI rendering model that directly handles reflectance, surface normal, and illumination. Our model thus gives a more accurate simulation of the physical imaging process and considers more imaging elements.

B. IHID Framework With DSM Cues

Local smooth prior on reflectance is a prior typically used for IHID, which says that neighboring pixels tend to share similar reflectance. To avoid global inconsistency in reflectance, we further introduce a nonlocal prior that utilizes DSM elevation information. By introducing DSM cues, our model can successfully eliminate the reflectance variance caused by the sunny and shady sides of buildings and is less sensitive to illumination changes. Moreover, the proposed cost function, which is based on the HSI rendering function and two priors, has a quadratic form and can be conveniently transformed into a linear system.

The rest of this article is organized as follows. Section II introduces the proposed HSI rendering model with DSM. The proposed IHID with the DSN cues model is given in Section III. The experimental results and analysis are reported in Section IV. Section V concludes this article.

II. HSI Rendering Model

It is essential to propose a proper HSI rendering model such that the scene properties can be recovered accurately. In this section, we show how to extend the typical HSI rendering model for HSIs and DSMs.

A. Pixelwise Form

The typical HSI rendering model is described as

$$H(\lambda) = R(\lambda) + S \tag{2}$$

where λ is the wavelength, $H(\lambda)$ and $R(\lambda)$, are the log-HSI, log reflectance, and component over each channel, respectively, and S is the log-shading component. It is a severely ill-posed problem and is solved by proposing priors on the distribution of reflectance and using the correlation between HSI channels. Nevertheless, it only solves a subproblem of the "intrinsic image" problem, as the shading component can be further decomposed into illumination and geometric components. More specifically, the shading component can be formulated as follows:

$$S = \log\left(\langle \mathbf{n}, \mathbf{v} \rangle\right) \tag{3}$$

where $\mathbf{n} = [N^x, N^y, N^z]^\intercal$ is the surface normal and $\mathbf{v} = [v^x, v^y, v^z]^\intercal$ is the direction of incident light. Directly formulating it in the log domain will involve the log of the dot product of two 3-D vectors, which is very inconvenient in practice. Hence, we apply the model of illumination proposed in [43] to generate log shading

$$S = \mathrm{SH}(\mathbf{n}, \mathbf{L}) \tag{4}$$

where $\mathbf{L} = [L_1, L_2, \ldots, L_9]^\intercal$ are first- and second-order spherical-harmonic illumination parameters and $\mathrm{SH}(\mathbf{n}, \mathbf{L})$ is

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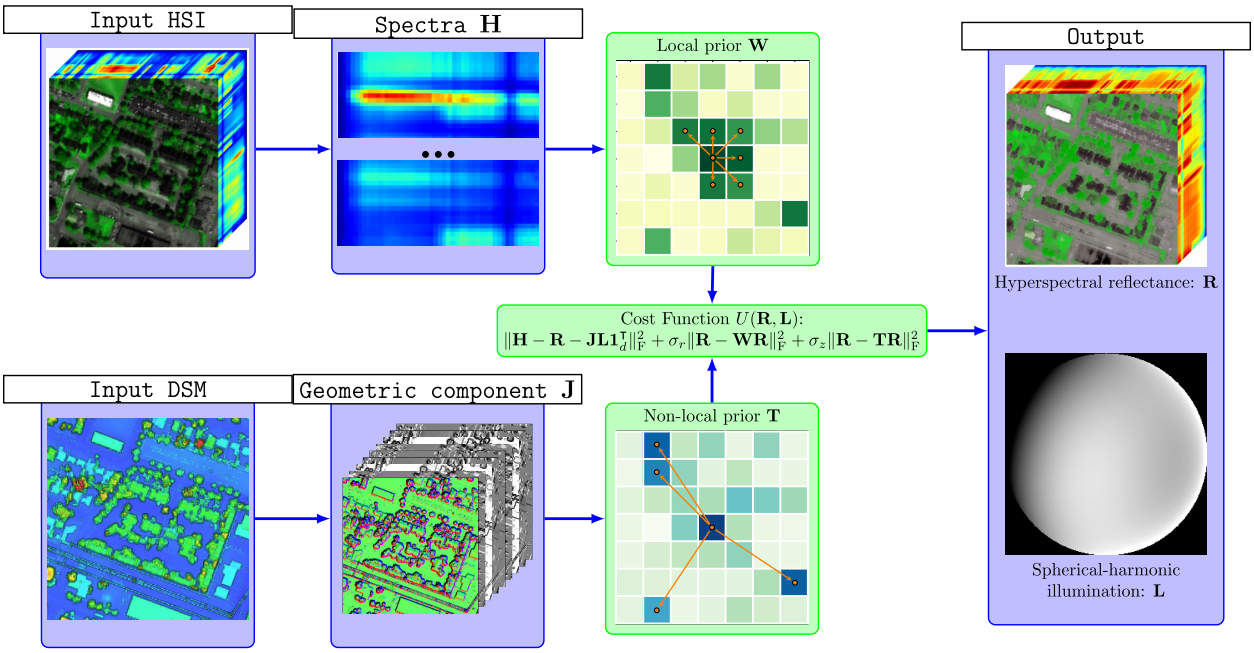

Fig. 1. Flowchart of the proposed method. Given input HSI and DSM, we first compute the spectral and geometric components; then, we introduce local prior and nonlocal prior to assure local and global consistency; Finally, we propose a quadratic cost function based on the HSI rendering model and priors and use an efficient conjugate gradient optimization method to obtain reflectance and spherical-harmonic illumination.

a spherical-harmonic model of illumination that generates log shading

$$\mathrm{SH}(\mathbf{n},\mathbf{L}) = [\mathbf{n};1]^\intercal \mathbf{M}[\mathbf{n};1] \tag{5}$$

$$\mathbf{M} = \begin{bmatrix} c_1L_9 & c_1L_5 & c_1L_8 & c_2L_4 \\ c_1L_5 & -c_1L_9 & c_1L_6 & c_2L_4 \\ c_1L_8 & c_1L_6 & c_3L_7 & c_2L_3 \\ c_2L_4 & c_2L_2 & c_2L_3 & c_4L_1 - c_5L_7 \end{bmatrix} \tag{6}$$

where $c1 = 0.429, c2 = 0.512, c3 = 0.743, c4 = 0.886$, and $c5 = 0.248$ are five constants.

Then, we rewrite (3) as

$$H(\lambda) = R(\lambda) + \mathrm{SH}(\mathbf{n},\mathbf{L}). \tag{7}$$

B. Matrix Form

We first write a log-HSI as a $u \times d$ matrix with columns representing pixels and rows representing log-spectral features over each wavelength, where u is the number of pixels and d is the number of channels. More specifically, channel k is denoted as $\mathbf{H}_k = [H_1(\lambda_k), H_2(\lambda_k), \ldots, H_u(\lambda_k)]^\intercal$, and the whole HSI is then denoted as $\mathbf{H} = [\mathbf{H}_1, \mathbf{H}_2, \ldots, \mathbf{H}_d]$. Similarly, let $\mathbf{R} = [\mathbf{R}_1, \mathbf{R}_2, \ldots, \mathbf{R}_d] \in u \times d$ be the log-reflectance image, where $\mathbf{R}_k = [R_1(\lambda_k), R_2(\lambda_k), \ldots, R_u(\lambda_k)]^\intercal$ is the channel k of the log-reflectance image. Then, we denote the DSM as a u-dimensional vector $\mathbf{Z} = [Z_1, Z_2, \ldots, Z_u]^\intercal$, where Z_k is the elevation at pixel k. With $\mathbf{Z}$, we can then compute the surface normal, $\mathbf{N} = [\mathbf{n}_1, \mathbf{n}_2, \ldots, \mathbf{n}_u]^\intercal$, where $\mathbf{N}$ is a $u \times 3$ vector and $\mathbf{n}_k = [N_k^x, N_k^y, N_k^z]^\intercal$. Note

$$\mathrm{SH}(\mathbf{n},\mathbf{L}) = [\mathbf{n}_k;1]^\intercal \mathbf{M}[\mathbf{n}_k;1] = \mathbf{J}_k^\intercal \mathbf{L} \tag{8}$$

$$\mathbf{J}_k = \left[c_4, 2c_2N_k^y, 2c_2N_k^x, c_2N_k^z, 2c_1N_k^xN_k^y, 2c_1N_k^yN_k^z \right. \\ \left. c_3N_k^zN_k^z - c_5, 2c_1N_k^xN_k^z, c_1\left(N_k^xN_k^x - N_k^yN_k^y\right)\right]^\intercal. \tag{9}$$

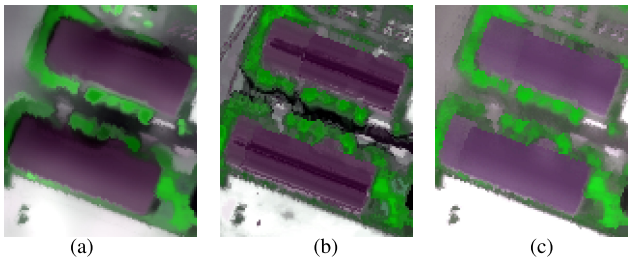

Fig. 2. Reflectance obtained by using (a) local prior only, (b) nonlocal prior only, and (c) both priors. We can see only by using both priors in conjunction dose. We obtain locally and globally consistent reflectance.

We then can define a geometric component, $\mathbf{J} = [\mathbf{J}_1, \mathbf{J}_2, \ldots, \mathbf{J}_u]^\intercal$.

Last, the matrix form of (7) is written as

$$\mathbf{H} = \mathbf{R} + \mathbf{JL1}_d^\intercal \tag{10}$$

where $\mathbf{1}_d$ is a d-dimensional vector with all 1's.

From (10), we can see that in an HSI, $\mathbf{H}$ is the interplay of material reflectance, $\mathbf{R}$, the geometric component, $\mathbf{J}$, and illumination, $\mathbf{L}$.

III. Proposed IHID With DSM Cues Model

We define the intrinsic scene decomposition with the DSM cues problem as follows: given an HSI, $\mathbf{H}$, and DSM, $\mathbf{Z}$, propose a model that can recover the hyperspectral reflectance $\mathbf{R}$ and illumination $\mathbf{L}$ based on (10). The overall flowchart is shown in Fig. 1. Note that we first use $\mathbf{Z}$ to compute the surface normal, $\mathbf{N}$, and then use $\mathbf{N}$ to compute $\mathbf{J}$.

A. Priors

This is a severely ill-posed problem, and we introduce two priors to make it solvable.

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1) Local Prior on Reflectance: The local prior on reflectance states that local neighboring pixels tend to share similar reflectance, that is to say, the log reflectance of pixel i can be expressed as a linear combination of the log reflectances of its neighboring pixels

$$\boldsymbol{\rho}_i = \sum_{j \in Q_i^l} \mathbf{W}_{ij} \boldsymbol{\rho}_j \tag{11}$$

where $\boldsymbol{\rho}_i = [R_i(\lambda_1), R_i(\lambda_2), \ldots, R_i(\lambda_d)]$ is the log reflectance of pixel i, $\boldsymbol{\rho}_j$ is the log reflectance of pixel j, Q_i^l is the set of neighboring pixels of pixel i, and $\mathbf{W}$ is a $u \times u$ sparse matrix whose nonzero element $\mathbf{W}_{ij}$ measures the similarity between pixels i and j.

L1-graph coding is used to calculate $\mathbf{W}_{ij}$

$$\min_{\alpha^i} \|\alpha^i\|_1 \quad \text{s.t.} \quad \mathcal{I}_i = \mathbf{D}^i \alpha^i \tag{12}$$

where $\mathcal{I}_i = [H_i(\lambda_1), H_i(\lambda_2), \ldots, H_i(\lambda_d)]$ is the log spectrum of pixel i, matrix $\mathbf{D}^i = \left[\mathcal{I}_1, \ldots, \mathcal{I}_{i-1}, \mathcal{I}_{i+1}, \ldots, \mathcal{I}_u, \mathbf{I}_d\right] \in R^{d \times (d+u-1)}$, and $\alpha^i \in R^{d+u-1}$ is the sparse coefficient. Then, set $\mathbf{W}_{ij} = \alpha_j^i$ if $i > j$ and $\mathbf{W}_{ij} = \alpha_{j-1}^i$ if $i < j$. L1-graph coding constructs a spare relation of given data and results in a spare $\mathbf{W}$. To speed up, we only use neighboring pixels in Q_i^l to construct $\mathbf{D}^i$ and set other elements to zeros.

The matrix form of (11) is then

$$\mathbf{R} = \mathbf{W}\mathbf{R}. \tag{13}$$

2) Nonlocal Prior on Reflectance: The nonlocal prior on reflectance is based on the observation that pixels with similar elevations tend to share similar reflectance. We first build a $(d+1)$-dimensional vector for each pixel as $[Z_i, H_i(\lambda_1), H_i(\lambda_2), \ldots, H_i(\lambda_d)]$, and we then find the global k-nearest neighbors Q_i^g using the Euclidean distance between the built feature vectors for each pixel $i = 1, 2, \ldots, u$. The nonlocal prior is

$$\boldsymbol{\rho}_i = \sum_{j \in Q_i^g} \mathbf{T}_{ij} \boldsymbol{\rho}_j \tag{14}$$

where $\mathbf{T}$ is a $u \times u$ sparse matrix whose nonzero element $\mathbf{T}_{ij}$ measures the similarity between pixels i and j. The matrix form of (14) is then

$$\mathbf{R} = \mathbf{T}\mathbf{R}. \tag{15}$$

The nonlocal prior makes use of the elevation information and can assure a globally consistent result.

Fig. 2 shows a demonstration of the effects of local and nonlocal priors on reflectance.

B. Optimization

The cost function we propose is based on the HSI rendering function and the two priors

$$U(\mathbf{R}, \mathbf{L}) = \left\|\mathbf{H} - \mathbf{R} - \mathbf{J}\mathbf{L}\mathbf{1}_d^\intercal\right\|_F^2 + \sigma_r \|\mathbf{R} - \mathbf{W}\mathbf{R}\|_F^2 + \sigma_z \|\mathbf{R} - \mathbf{T}\mathbf{R}\|_F^2 \tag{16}$$

where σ_r and σ_z are weights of the two priors. $\|\mathbf{M}\|_F$ denotes the Frobenius norm of matrix $\mathbf{M}$. In this article, σ_r and σ_z are empirically set to 50 and 10, respectively.

Since U is a quadratic function, the global optimum lies at the root of its first partial derivatives, which can be calculated as follows:

$$\frac{1}{2}\frac{\partial U}{\partial \mathbf{R}} = \mathbf{G}\mathbf{R} + \mathbf{J}\mathbf{L}\mathbf{1}_d^\intercal - \mathbf{H} \tag{17}$$

$$\mathbf{G} = \mathbf{I}_u + \sigma_r(\mathbf{I}_u - \mathbf{W}^\intercal)(\mathbf{I}_u - \mathbf{W}) + \sigma_z(\mathbf{I}_u - \mathbf{T}^\intercal)(\mathbf{I}_u - \mathbf{T}) \tag{18}$$

$$\frac{1}{2}\frac{\partial U}{\partial \mathbf{L}} = d\mathbf{J}^\intercal\mathbf{J}\mathbf{L} - \mathbf{J}(\mathbf{H} - \mathbf{R})\mathbf{1}_d \tag{19}$$

where $\mathbf{I}_u$ is a $u \times u$ unit matrix.

Let $\partial U/\partial \mathbf{R} = 0$ and $\partial U/\partial \mathbf{L} = 0$, and we have

$$\left[\begin{array}{cccc:c} \mathbf{G} & & & & \mathbf{J} \\ & \mathbf{G} & & & \mathbf{J} \\ & & \ddots & & \vdots \\ & & & \mathbf{G} & \mathbf{J} \\ \hdashline \frac{1}{d}\mathbf{J}^\intercal & \frac{1}{d}\mathbf{J}^\intercal & \cdots & \frac{1}{d}\mathbf{J}^\intercal & \mathbf{J}^\intercal\mathbf{J} \end{array}\right] \left[\begin{array}{c} \mathbf{R}_1 \\ \mathbf{R}_2 \\ \vdots \\ \mathbf{R}_d \\ \mathbf{L} \end{array}\right] = \left[\begin{array}{c} \mathbf{H}_1 \\ \mathbf{H}_2 \\ \vdots \\ \mathbf{H}_d \\ \frac{1}{d}\mathbf{J}^\intercal\mathbf{H} \end{array}\right]. \tag{20}$$

We refer interested readers to the appendix for how (17), (19), and (20) are derived in detail.

Equation (20) is a linear system with the form of $\mathbf{A}\mathbf{x} = \mathbf{b}$, where $\mathbf{A}$ is a $(ud+9) \times (ud+9)$ square matrix, $\mathbf{x}$ and $\mathbf{b}$ are $(ud+9)$-dimensional vectors, and $\mathbf{x}$ is the unknown. Since $\mathbf{A}$ is a sparse matrix, (20) can be efficiently solved by a conjugate gradient method. Solving (20), we then obtain $\mathbf{R} = [\mathbf{R}_1, \mathbf{R}_2, \ldots, \mathbf{R}_d]$ and $\mathbf{L}$.

IV. Experimental Results and Discussion

In this section, we report experiments on both synthetic and real data to evaluate the effectiveness of the proposed IHID with the DSM cues model.

A. Experiments on Pseudosynthetic Data

1) Data Description: It is rather challenging to produce ground-truth hyperspectral reflectance and illumination for real natural scenes. To quantitatively evaluate the performance of the proposed method, we compose pseudosynthetic scenes based on the "grss_dfc_2018" dataset, which is used for the 2018 IEEE GRSS Data Fusion Contest [44]–[46]. Our idea is first to generate an intrinsic hyperspectral point cloud that we can deem as ground truth, based on which we can synthesize a series of HSIs on different illuminations. More specifically, we take a subscene from the "grss_dfc_2018" dataset with the LiDAR point cloud and corresponding HSI. We first run the HSI decomposition algorithm in [36] on HSI to obtain a 2-D reflectance image, which still suffers from spectral mixing and shadow effects. We manually segment the point cloud into different objects and manually assign reflectance from the 2-D reflectance image for each object (avoiding using reflectance from image boundaries and shadow regions). With the 3-D terrain and 3-D surface reflectances from the generated intrinsic hyperspectral point cloud, we then synthesize three HSIs with different illuminations with a discrete Monte Carlo approach. The DSM and ground-truth hyperspectral reflectance degenerate from the intrinsic hyperspectral point cloud. The data are shown in Fig. 3, which includes the

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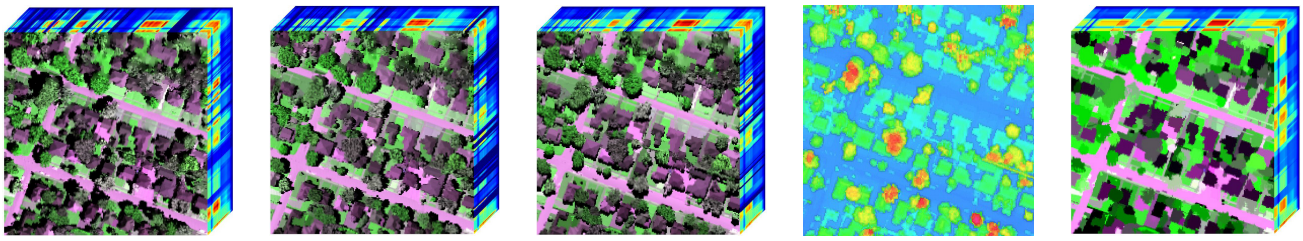

(a) (b) (c) (d) (e)

Fig. 3. Pseudosynthetic HSIs and DSM with ground-truth hyperspectral reflectance. (a) HSI_light1, (b) HSI_light2, and (c) HS_light3 are the three synthetic HSIs of different light conditions. (d) DSM. (e) Ground-truth hyperspectral reflectance.

following five subdata: three HSI cubes synthesized under different light conditions, the DSM, and the ground-truth hyperspectral reflectance of the scene.

2) Results and Analysis: We apply the proposed IHID with DSM cues method on the three HSIs with different light conditions and the DSM. The state-of-the-art alternative methods we compare are: 1) Jin *et al.*'s method [36], a noniterative intrinsic decomposition method designed for HSI; 2) Kang's method [37], which first divideed the spectral bands of an HSI into several subgroups and then conducted IID separately; and 3) Jeon *et al.*'s method [31], which first separated an RGB image into cartoon and texture components and then conducted IID with surface normals. Note that the three alternative methods are not originally designed for HSIs and DSM data, and some modifications are made to fit the new data as follows. For the methods proposed by Jin *et al.* [36] and Kang *et al.* [37], DSM information is added to compute the similarity between two pixels; for Jeon *et al.*'s method [31], the HSI is divided into multiple three-channel subimages to perform the algorithm, and the results are merged to form a hyperspectral reflectance. The false-color images of the obtained hyperspectral reflectance and shading component obtained by different methods are shown in Fig. 4. Note that the spherical-harmonic illumination estimated by the proposed method is rendered into a shading component for more precise visualization. From Fig. 4, we can see that our output reflectance and shading images look better than those of the state-of-the-art intrinsic decomposition methods. More specifically, our model tends to give illumination-invariant reflectance images and the shadow effects are considerably reduced, which shows that it is more robust to illumination changes. Our shading component contains more details of the geometric structures of the ground surface.

The quantitative reflectance mean squared error (RMSE) of different methods are shownin Fig. 5. The RMSE error metric is computed as

$$R_{\text{MSE}}(\hat{\mathbf{R}}, \mathbf{R}^*) = \sum_{\lambda} \min_{\alpha} \|\alpha \hat{\mathbf{R}}(\lambda) - \mathbf{R}^*(\lambda)\|_2^2 \qquad (21)$$

where $\mathbf{R}^*$ is the ground-truth hyperspectral reflectance and $\hat{\mathbf{R}}$ is the recovered hyperspectral reflectance.

Fig. 5 shows that the proposed method has the slightest error and is not sensitive to light conditions. For other alternative methods, data scene 1 is the most challenging data and has reported much higher errors on this dataset than on the other two datasets. In contrast, our method achieves satisfactory results on all three datasets, which demonstrates its resoluteness. We also find that Jeon [31] has the worst performance, which shows that a typical IID method for RGB-D works poorly for HSIs. We think that the reason is that HSIs usually have much lower spatial resolution than natural RGB images; as a result, HSIs provide poorer shading cues than RGB images. In this case, RGB IID methods that rely on the quality of shading cues work poorly.

TABLE I
COMPUTING TIME OF THE DIFFERENT METHODS

Scene	Ours	Jin. [36]	Kang. [37]	Jeon. [31]
HSI_light1	$3m8s$	$2m46s$	$1m15s$	$1m14s$
HSI_light2	$3m06s$	$2m50s$	$1m10s$	$1m15s$
HSI_light3	$3m12s$	$2m48s$	$1m17s$	$1m08s$

3) Parameter Influence: In this section, we study the influence of parameter changes on our method. Fig. 6 shows the RMSE with changes in neighborhood sizes, where $|Q^l|$ is the neighborhood size of the local prior and $|Q^s|$ is the neighborhood size of nonlocal prior. Overall, the flat plots show that neighborhood sizes do not greatly influence the results very much (the standard deviation is less than 0.002). Combined with the fact that increasing neighborhood size increases the nonzero elements of $\mathbf{G}$ and, thus, the complexity, it is better to choose a smaller neighborhood size. In Fig. 7, we can see the influence of the weights of the two priors, σ_r and σ_z. The RMSE of using only one prior (when $\sigma_r = 0$ or $\sigma_z = 0$) is much larger than that of using both priors, which demonstrates the importance of both priors. We can see that our model achieves the best performance for $\sigma_r \in [20, 100]$ and $\sigma_z \in [10, 50]$. The RMSE is slightly worse for $\sigma_z > 50$ although the variance is very small.

4) Complexity: The most time-consuming process of our method is to optimize the cost function, which has a time complexity of $\mathcal{O}(m\sqrt{\kappa})$ and a space complexity of $\mathcal{O}(m)$, where m is the number of nonzero elements in $\mathbf{A}$ and κ is the spectral condition number of $\mathbf{A}$. In our case, $m \in \mathcal{O}(ud)$. The computing times of the different methods are reported in Table I. The tests are run using an Intel Core i7-6800K processor and 128 GB of RAM. Our method is among the slowest methods, as it needs to handle more scene properties than other methods.

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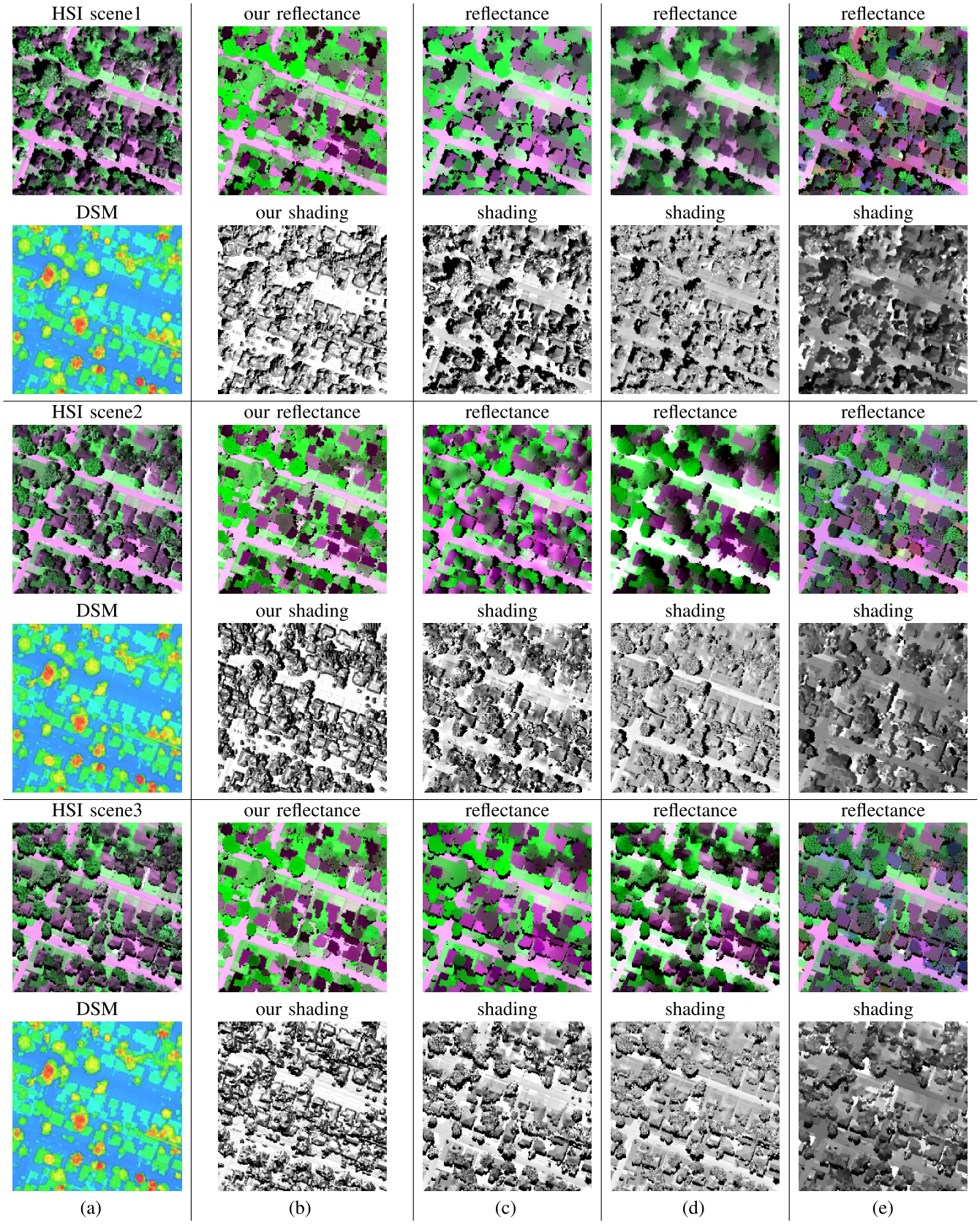

Fig. 4. Results on synthetic datasets. (a) Input HSI (false-color images) and DSM. (b) Results (false-color images of hyperspectral reflectance and the shading image) of the proposed model. Results of alternative methods: (c) Jin [36], (d) Kang [37], and (e) Jeon [31].

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TABLE II
CLASSIFICATION ACCURACY (%) OF VARIOUS FEATURES ON THE MUUFL GULFPORT DATA

No.	MUUFL GULFPORT						
	Categories	Num. of samples	Ori.	Ours	Jin. [36]	Kang. [37]	Jeon. [31]
C1	Trees	23246	96.79 ± 0.51	95.91 ± 0.52	95.97 ± 0.65	$\mathbf{97.58 \pm 0.53}$	97.34 ± 0.43
C2	Mostly grass	4270	70.91 ± 3.24	85.57 ± 2.60	$\mathbf{87.51 \pm 2.85}$	83.56 ± 2.26	78.73 ± 1.57
C3	Mixed ground surface	6882	76.20 ± 1.49	$\mathbf{88.49 \pm 1.65}$	88.01 ± 1.35	87.10 ± 1.56	83.73 ± 1.80
C4	Dirt/sand	1826	64.22 ± 5.24	$\mathbf{89.79 \pm 2.20}$	82.83 ± 5.44	83.11 ± 3.50	68.20 ± 3.60
C5	Road	6687	90.80 ± 1.71	94.30 ± 0.91	90.82 ± 1.74	$\mathbf{94.42 \pm 1.00}$	92.56 ± 1.15
C6	Water	466	85.54 ± 4.03	$\mathbf{88.08 \pm 3.48}$	85.20 ± 2.46	24.16 ± 7.39	85.01 ± 3.90
C7	Buildings	6240	90.27 ± 1.84	$\mathbf{93.61 \pm 1.02}$	93.60 ± 0.97	93.31 ± 1.42	91.23 ± 1.71
C8	Sidewalk	1385	40.75 ± 3.71	52.30 ± 4.81	46.73 ± 4.60	$\mathbf{57.80 \pm 4.87}$	45.17 ± 4.15
C9	Yellow curb	183	13.83 ± 4.22	$\mathbf{39.23 \pm 16.14}$	38.28 ± 11.89	21.12 ± 7.08	9.97 ± 4.68
C10	Cloth panels	629	41.39 ± 8.27	$\mathbf{76.65 \pm 12.12}$	64.13 ± 11.63	60.89 ± 10.11	49.11 ± 10.41
	OA		86.97 ± 0.37	$\mathbf{92.35 \pm 0.37}$	90.99 ± 0.44	91.37 ± 0.31	89.50 ± 0.33
	AA		67.07 ± 1.32	$\mathbf{80.29 \pm 1.82}$	77.31 ± 1.81	70.31 ± 1.61	70.10 ± 1.42
	Kappa		82.11 ± 0.52	$\mathbf{89.58 \pm 0.51}$	87.73 ± 0.60	88.20 ± 0.43	85.51 ± 0.49

TABLE III
CLASSIFICATION ACCURACY (%) OF VARIOUS FEATURES ON THE HU 2018 DATA

No.	HU2018						
	Categories	Num. of samples	Ori.	Ours	Jin. [36]	Kang. [37]	Jeon. [31]
C1	Healthy grass	399	72.69 ± 7.87	67.14 ± 9.01	55.69 ± 10.11	56.35 ± 7.95	$\mathbf{73.65 \pm 9.05}$
C2	Stressed grass	4694	92.16 ± 1.73	92.96 ± 1.78	91.36 ± 2.34	89.09 ± 1.26	$\mathbf{94.23 \pm 1.62}$
C3	Evergreen trees	1818	83.20 ± 3.80	$\mathbf{89.59 \pm 3.75}$	87.78 ± 2.90	77.36 ± 4.12	84.69 ± 3.52
C4	Deciduous trees	802	63.18 ± 6.31	$\mathbf{72.60 \pm 5.86}$	71.83 ± 7.29	56.08 ± 7.43	70.41 ± 6.30
C5	Residential buildings	31709	87.63 ± 0.69	$\mathbf{93.33 \pm 0.91}$	92.18 ± 0.61	93.19 ± 0.65	91.52 ± 0.61
C6	Non-residential buildings	26037	86.73 ± 0.74	90.61 ± 0.86	92.08 ± 0.66	$\mathbf{94.14 \pm 0.56}$	91.01 ± 0.73
C7	Roads	17597	68.71 ± 1.10	79.94 ± 1.07	$\mathbf{80.75 \pm 1.01}$	78.75 ± 1.24	75.61 ± 1.13
C8	Sidewalks	1560	38.71 ± 4.33	$\mathbf{47.58 \pm 5.20}$	46.00 ± 3.06	32.38 ± 3.97	44.77 ± 5.38
C9	Crosswalks	66	35.98 ± 18.33	$\mathbf{51.67 \pm 16.53}$	37.80 ± 13.76	41.52 ± 13.02	40.00 ± 16.59
C10	Major thoroughfares	20504	71.58 ± 1.14	$\mathbf{87.95 \pm 1.49}$	86.23 ± 1.62	84.45 ± 1.13	79.01 ± 1.46
C11	Paved parking lots	3490	87.54 ± 2.38	$\mathbf{97.99 \pm 0.93}$	96.70 ± 1.84	97.09 ± 2.48	95.63 ± 1.42
C12	Unpaved parking lots	146	86.27 ± 11.73	$\mathbf{90.58 \pm 5.05}$	83.60 ± 10.58	64.93 ± 12.74	85.51 ± 14.88
C13	Cars	2625	62.19 ± 3.09	$\mathbf{91.80 \pm 2.45}$	71.16 ± 3.57	78.56 ± 3.01	63.71 ± 2.23
	OA		80.05 ± 0.31	$\mathbf{88.71 \pm 0.41}$	87.83 ± 0.46	87.55 ± 0.28	85.15 ± 0.37
	AA		72.04 ± 2.11	$\mathbf{81.06 \pm 1.50}$	76.40 ± 1.65	72.61 ± 1.29	76.13 ± 1.82
	Kappa		75.11 ± 0.39	$\mathbf{85.93 \pm 0.51}$	84.82 ± 0.57	84.40 ± 0.36	81.45 ± 0.45

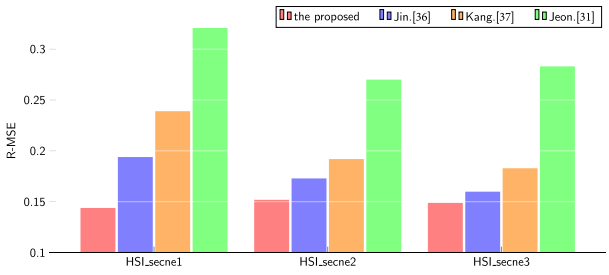

Fig. 5. RMSE of different methods on the three HSI datasets.

B. Experiments on Real Data

More results on the two real datasets are given in this section. As ground-truth reflectance of a real remote sensing scene is hardly accessible, we use visual perception and downstream classification performance to evaluate the reflectance recovery accuracy indirectly.

1) Data Description:

1) *MUUFL Gulfport:* This HSI was collected in November 2010 using a Gemini LiDAR and CASI-1500 flown in a single plane simultaneously over the University of Southern Mississippi Gulf Park Campus [47]. The data collection contains coregistered HSI and LiDAR DSM. The MUUFL Gulfport dataset campus 1 scene we used in this article contains 325 × 337 pixels across 72 bands. After initial screening, some areas and several bands were removed due to noise, resulting in a new image with the size of 325 × 220 × 64 [48]. Original data distributed in 11 categories; as the "Shadow" category is not a real land cover, we removed this category and remained the other ten categories. Fig. 8 shows the false-composite image and reference information of the MUUFL Gulfport dataset.
2) *HU 2018:* The data we use are scene 272056_3289689 of the "grss_dfc_2018" dataset, which includes an HSI

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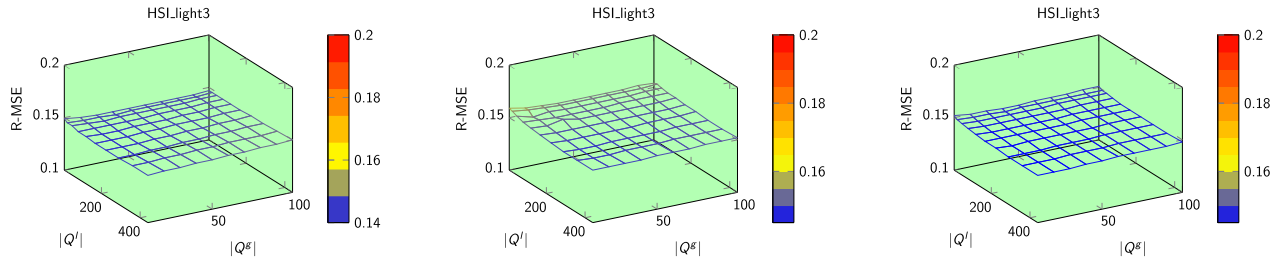

Fig. 6. RMSE with different neighborhood sizes. The results of (a) data HSI_scene1, (b) HSI_scene2, and (c) HSI_scene3.

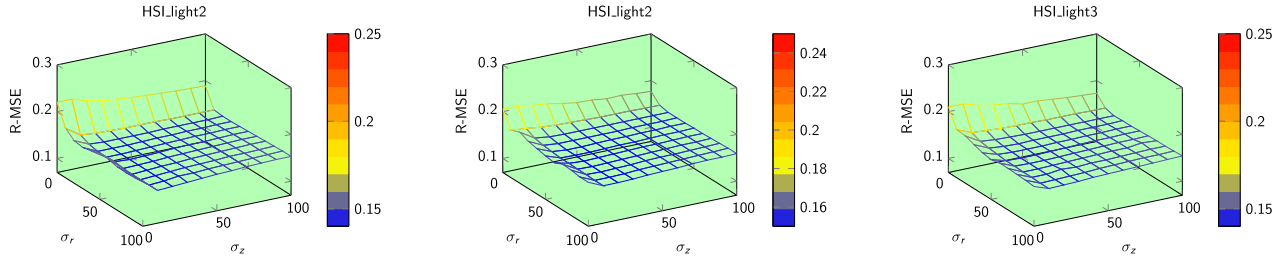

Fig. 7. RMSE with the change of weights of the two priors, σ_r and σ_z. The results of (a) data HSI_scene1, (b) HSI_scene2, and (c) HSI_scene3.

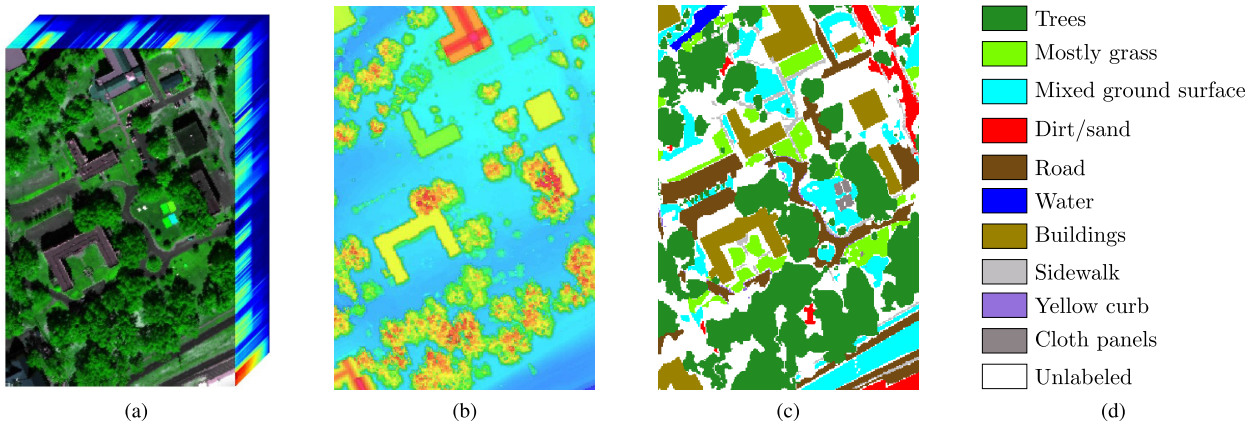

Fig. 8. MUUFL GUFLPORT dataset. (a) HSI. (b) LiDAR DSM. (c) Ground-truth map. (d) Legend.

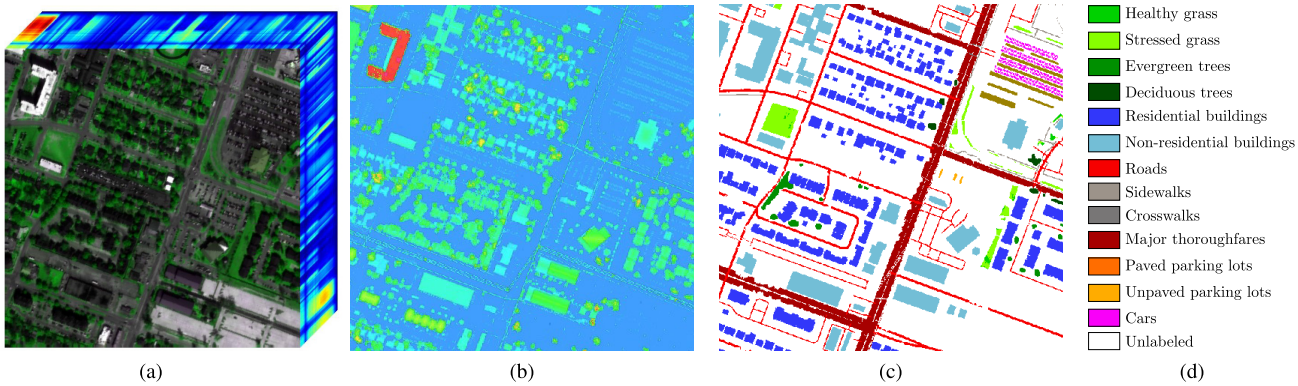

Fig. 9. Scene 272056_3289689 of "grss_dfc_2018" dataset. (a) HSI. (b) LiDAR DSM. (c) Ground-truth map. (d) Legend.

with a resolution of 601 × 596, 48 spectral bands, and a LiDAR DSM. We have 13 categories and detailed data information given in Fig. 9.

2) Visual Results: Fig. 10 shows the recovered hyperspectral reflectance component and the shading component. Similar to synthetic data, the spherical-harmonic illumination

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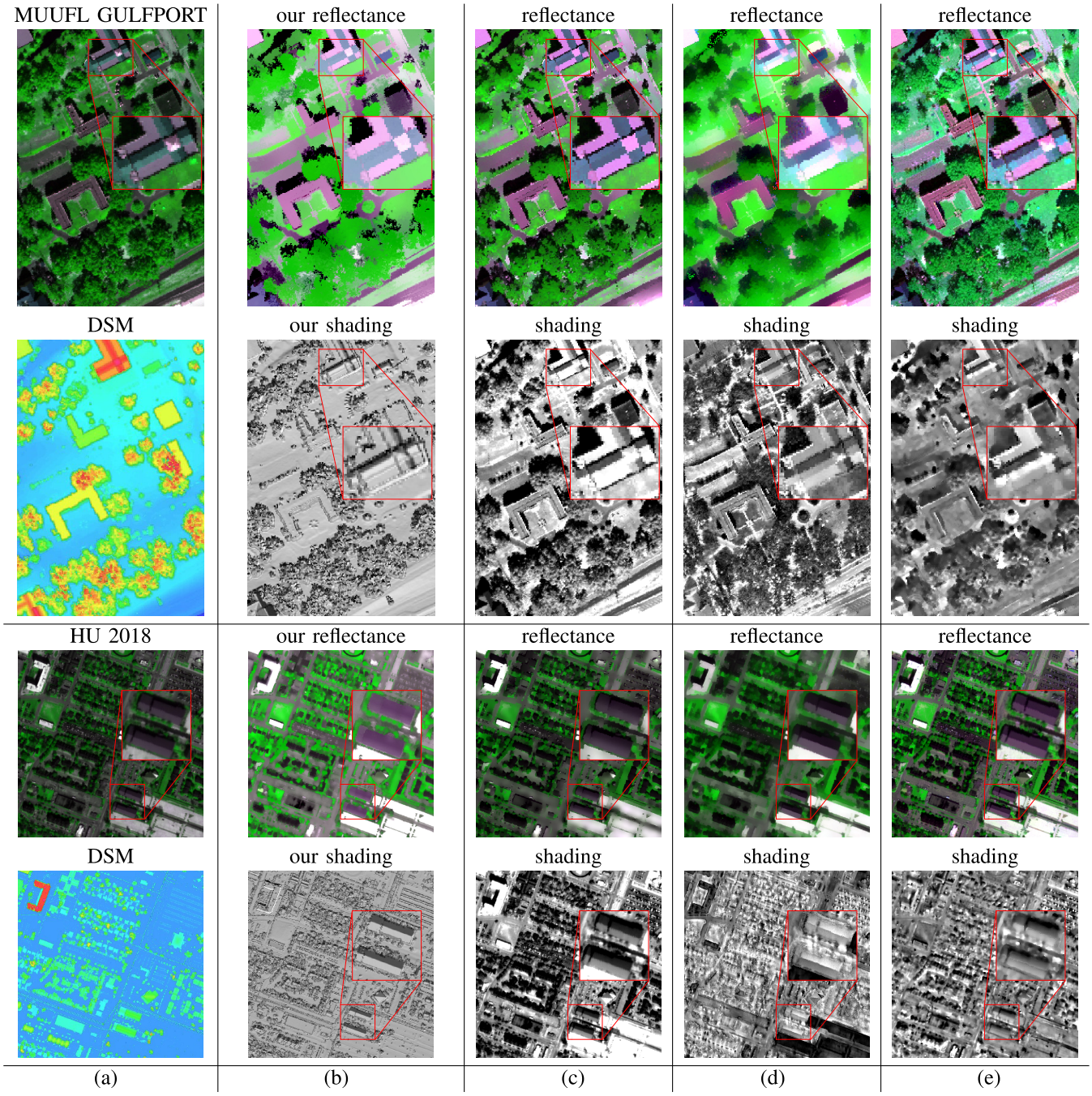

Fig. 10. Results on real datasets. (a) Input HSI (false-color images) and DSM. (b) Results (false-color images of hyperspectral reflectance and the shading image) of the proposed model. Results of alternative methods: (c) Jin [36], (d) Kang [37], and (e) Jeon [31].

estimated by our method is rendered into shading with the geometric component for a more intuitive visual effect. As we can see from the red boxes in Fig. 10 where certain areas are marked and magnified, the proposed method has two major strengths: 1) our method succeeds to recover uniform reflectance in the sunny side and the shady side of a single building which is expected to share similar reflectance; 2) our method can considerably eliminate shadows near buildings, given that the shadow area is not too large. We also observe that the HU 2018 dataset is especially challenging for IID. Methods, such as Jin [36], and Kang [37], depend on a threshold of gradient change to determine whether two pixels share similar reflectance; this works poorly for HU 2018 data, and in a way, the reflectances of some areas with different materials appear to be oversmoothed, while those of other areas fail to be recovered correctly. In contrast, our method can successfully handle this issue due to our global priors on reflectance. Moreover, benefitting from the spherical-harmonic illumination model, our shading components also seem to present more meaningful geometric structures than alternative methods.

3) Classification Performance: As reflectance is dependent only on the material properties and irrelevant to environmental imaging elements such as illumination and geometry structures, it is expected to benefit other HSI processing tasks

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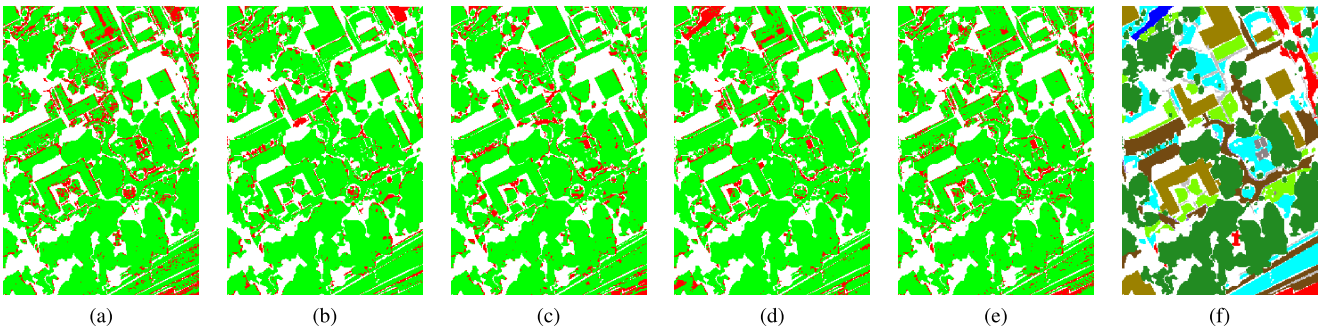

Fig. 11. Classification maps of different methods on MUUFL GULFPORT data. (a) Ori. (b) Ours. (c) Jin [36]. (d) Kang [37]. (e) Jeon [31]. (f) Ground truth.

Fig. 12. Classification maps of different methods on HU 2018 data. (a) Ori. (b) Ours. (c) Jin [36]. (d) Kang [37]. (e) Jeon [31]. (f) Ground truth.

such as HSI classification. Hence, the boost that the recovered reflectance brings can also be used as an indirect evaluation index. Since our goal is to compare the effectiveness of different IHID methods by demonstrating the discriminating power of hyperspectral reflectance, we employ a support vector machine (SVM) as a classifier other than using other more complicated classifiers. SVM is a widely used classifier with sound theory and robust performance. We have the following five feature settings as inputs.

1) *Ori.:* Concatenated original hyperspectral spectral features and DSM feature.
2) *Ours:* Concatenated hyperspectral reflectance features recovered by our method and DSM features.
3) *Jin [36]:* Concatenated hyperspectral reflectance features recovered by Jin [36] and DSM features.
4) *Kang [37]:* Concatenated hyperspectral reflectance features recovered by Kang [37] and DSM features.
5) *Jeon [31]:* Concatenated hyperspectral reflectance features recovered by Jeon [31] and DSM features.

For each dataset, 2% of labeled samples are randomly chosen as training samples and the remaining as test samples. All the results reported are the mean and standard deviation

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of 20 repeated experiments. Conclusion was drawn from analyzing the classification accuracy assessed with the following measures:

1) an overall accuracy (OA), which is the number if correctly classified samples out of the total number of the tested samples in percentage;
2) an average accuracy (AA), which is the mean of the percentage of correctly labeled pixels for each class
3) a kappa coefficient of agreement, which is the percentage of agreement corrected by the amount of that could be excerpted due to chance alone.

The results on MUUFL GULFPORT and HU 2018 data are summarized in Tables II and III, respectively, from which we can see that our method achieves the best results on both datasets. Compared with the original spectral features, our method boosts the OA, AA, and Kappa by a large margin, i.e., an average approximately 7% boost for OA and Kappa, 11% for AA. Compared with other IHID methods, our method slightly outperforms these methods in terms of OA and Kappa while performing better for various categories, which are challenging to classify and thus boost AA by a larger margin. The difficulty of classifying these categories is caused by the following two reasons: 1) the limited number of existing samples, e.g., "Yellow curb" and "Cloth panels," in MUUFL data and "Crosswalks" and "Unpaved parking lots," and 2) small objects, e.g., "Cars," in HU data. As IID methods work by guessing which pixels share similar reflectance according to spatial cues, an aggressive guess will cause an oversmoothed reflectance image and performance worse for categories with limited samples or small objects. Our method, however, benefiting from the introduced DSM geometric cues, has a much better performance in such a case. The classification maps of different methods are given in Figs. 11 and 12.

V. Conclusion

In this article, we have presented an IHID with DSM cues model, which can effectively recover hyperspectral reflectance and environmental illumination from HSIs and DSM data. By using HSI alone, it can be difficult to handle the sunny and shady sides of the same materials and is also sensitive to illumination changes. By introducing DSM cues to IHID, we can further derive the environmental illumination of the scene and thus obtain a more illumination-invariant result. Experiments on both synthetic and real data demonstrate that our method outperforms state-of-the-art IHID methods in terms of recovery error and classification ac accuracy. As our method can reduce the effects of shadows to a certain degree, especially for shadows of small areas, it cannot eliminate large shadows of the scene. In future research, we will design a model that can handle the shadow problem.

Appendix

Proofs of the Proposed Propositions

In this section, the detailed proofs of (17), (19), and (20) are given for interested readers.

First, we recall that (17) and (19) are the first partial derivatives of the cost function (16)

$$U(\mathbf{R},\mathbf{L}) = \left\|\mathbf{H}-\mathbf{R}-\mathbf{J}\mathbf{L}\mathbf{1}_d^\intercal\right\|_F^2 + \sigma_r\|\mathbf{R}-\mathbf{W}\mathbf{R}\|_F^2 + \sigma_z\|\mathbf{R}-\mathbf{T}\mathbf{R}\|_F^2.$$

The cost function (16) is a quadratic function, and its partial derivative is thus a linear polynomial. Though it has a simple linear polynomial form, it is not trivial to derive it item by item. Here, we introduce some convenient matrix derivative formulas.

Theorem 1: If $\mathbf{X}$ is an $m \times n$ matrix and $\mathbf{A}$ is an $n \times m$ matrix, we have

$$\frac{\partial \operatorname{Tr}(\mathbf{A}\mathbf{X})}{\partial \mathbf{X}} = \mathbf{A}^\intercal \tag{22}$$

where $\operatorname{Tr}(\cdot)$ is the trace of a matrix.

Proof: The following condition holds.

$$\frac{\partial \operatorname{Tr}(\mathbf{A}\mathbf{X})}{\partial \mathbf{X}_{ij}} = \frac{\partial \sum_{s=1}^{n}\sum_{t=1}^{m}(\mathbf{A}_{st}\mathbf{X}_{ts})}{\partial \mathbf{X}_{ij}} = \mathbf{A}_{ji}. \tag{23}$$

As (23) holds for any i and j, we have (22). ∎

Theorem 2: If $\mathbf{X}$ is an $m \times n$ matrix, $\mathbf{A}$ is an $m \times m$ matrix, and $\mathbf{B}$ is an $n \times n$ matrix, we have

$$\frac{1}{2}\frac{\partial \operatorname{Tr}(\mathbf{A}\mathbf{X}\mathbf{B}\mathbf{X}^\intercal)}{\partial \mathbf{X}} = \mathbf{A}^\intercal\mathbf{X}\mathbf{B}^\intercal + \mathbf{A}\mathbf{X}\mathbf{B} \tag{24}$$

where $\operatorname{Tr}(\cdot)$ is the trace of a matrix.

Proof: First, we recall following properties of matrix trace: 1) $\operatorname{Tr}(\mathbf{C}+\mathbf{D}) = \operatorname{Tr}(\mathbf{C}) + \operatorname{Tr}(\mathbf{D})$; 2) $\operatorname{Tr}(\mathbf{C}) = \operatorname{Tr}(\mathbf{C}^\intercal)$; and 3) $\operatorname{Tr}(\mathbf{C}\mathbf{D}) = \operatorname{Tr}(\mathbf{D}\mathbf{C})$.

Then, let us focus on the derivative of $\operatorname{Tr}(\mathbf{A}\mathbf{X}\mathbf{B}\mathbf{X}^\intercal)$ with respect to a single element of $\mathbf{X}$, say $\mathbf{X}_{ij}$, and we have

$$\begin{aligned}\frac{\partial \operatorname{Tr}(\mathbf{A}\mathbf{X}\mathbf{B}\mathbf{X}^\intercal)}{\partial \mathbf{X}_{ij}} &= \operatorname{Tr}\left(\frac{\partial \mathbf{A}\mathbf{X}\mathbf{B}\mathbf{X}^\intercal}{\partial \mathbf{X}_{ij}}\right)\\ &= \operatorname{Tr}\left(\frac{\partial \mathbf{A}\mathbf{X}}{\partial \mathbf{X}_{ij}}\mathbf{B}\mathbf{X}^\intercal\right) + \operatorname{Tr}\left(\mathbf{A}\mathbf{X}\frac{\partial \mathbf{B}\mathbf{X}^\intercal}{\partial \mathbf{X}_{ij}}\right)\\ &= \operatorname{Tr}\left(\left[\frac{\partial \mathbf{A}\mathbf{X}}{\partial \mathbf{X}_{ij}}\right]^\intercal\mathbf{X}\mathbf{B}^\intercal\right) + \operatorname{Tr}\left(\mathbf{A}\mathbf{X}\frac{\partial \mathbf{B}\mathbf{X}^\intercal}{\partial \mathbf{X}_{ij}}\right)\\ &= \sum_k \mathbf{A}_{ki}\left[\mathbf{X}\mathbf{B}^\intercal\right]_{kj} + \sum_k [\mathbf{A}\mathbf{X}]_{ik}\mathbf{B}_{kj}\\ &= \left[\mathbf{A}^\intercal\mathbf{X}\mathbf{B}^\intercal\right]_{ij} + [\mathbf{A}\mathbf{X}\mathbf{B}]_{ij}.\end{aligned} \tag{25}$$

Since (25) holds for any i and j, we have

$$\frac{\partial \operatorname{Tr}(\mathbf{A}\mathbf{X}\mathbf{B}\mathbf{X}^\intercal)}{\partial \mathbf{X}} = \mathbf{A}^\intercal\mathbf{X}\mathbf{B}^\intercal + \mathbf{A}\mathbf{X}\mathbf{B}. \tag{26}$$

∎

Theorem 3: If $\mathbf{X}$ is an $m \times n$ matrix, $\mathbf{A}$ is an $s \times m$ matrix, $\mathbf{B}$ is an $n \times t$ matrix, and $\mathbf{Y}$ is an $s \times t$ matrix, then we have

$$\frac{1}{2}\frac{\partial \|\mathbf{Y}-\mathbf{A}\mathbf{X}\mathbf{B}\|_F^2}{\partial \mathbf{X}} = \mathbf{A}^\intercal\mathbf{A}\mathbf{X}\mathbf{B}\mathbf{B}^\intercal - \mathbf{A}^\intercal\mathbf{Y}\mathbf{B}^\intercal. \tag{27}$$

Proof: Since $||M||_F = (\operatorname{Tr}(M^\intercal M))^{1/2}$, we have

$$\begin{aligned}\partial\|\mathbf{Y}-\mathbf{A}\mathbf{X}\mathbf{B}\|_F^2 &= \operatorname{Tr}((\mathbf{Y}-\mathbf{A}\mathbf{X}\mathbf{B})^\intercal(\mathbf{Y}-\mathbf{A}\mathbf{X}\mathbf{B}))\\ &= \operatorname{Tr}(\mathbf{Y}^\intercal\mathbf{Y}) - 2\operatorname{Tr}(\mathbf{B}\mathbf{Y}^\intercal\mathbf{A}\mathbf{X})\\ &\quad + \operatorname{Tr}(\mathbf{A}^\intercal\mathbf{A}\mathbf{X}\mathbf{B}\mathbf{B}^\intercal\mathbf{X}^\intercal).\end{aligned} \tag{28}$$

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Then,

$$\frac{1}{2}\frac{\partial\|\mathbf{Y}-\mathbf{AXB}\|_F^2}{\partial\mathbf{X}}=\frac{1}{2}\frac{\partial\,\mathrm{Tr}(\mathbf{A}^\intercal\mathbf{AXBB}^\intercal\mathbf{X}^\intercal)}{\partial\mathbf{X}}-\frac{\partial\,\mathrm{Tr}(\mathbf{BY}^\intercal\mathbf{AX})}{\partial\mathbf{X}}. \tag{29}$$

Applying (22) and (24) to (29), we immediately get (27). ■

Applying (27) to (16), we then have

$$\begin{aligned}\frac{1}{2}\frac{\partial U}{\partial\mathbf{R}}&=\left(\mathbf{R}-\left(\mathbf{H}-\mathbf{JL1}_d^\intercal\right)\right)\\&\quad+\sigma_r(\mathbf{I}-\mathbf{W}^\intercal)(\mathbf{I}-\mathbf{W})\mathbf{R}+\sigma_z(\mathbf{I}-\mathbf{T}^\intercal)(\mathbf{I}-\mathbf{T})\mathbf{R}\\&=\mathbf{GR}+\mathbf{JL1}_d^\intercal-\mathbf{H}\end{aligned} \tag{30}$$

$$\begin{aligned}\mathbf{G}&=\mathbf{I}_u+\sigma_r(\mathbf{I}_u-\mathbf{W}^\intercal)(\mathbf{I}_u-\mathbf{W})\\&\quad+\sigma_z(\mathbf{I}_u-\mathbf{T}^\intercal)(\mathbf{I}_u-\mathbf{T})\end{aligned} \tag{31}$$

$$\frac{1}{2}\frac{\partial U}{\partial\mathbf{L}}=\mathbf{J}^\intercal\mathbf{JL1}_d^\intercal\mathbf{1}_d-\mathbf{J}^\intercal(\mathbf{H}-\mathbf{R})\mathbf{1}_d \tag{32}$$

$$=d\mathbf{J}^\intercal\mathbf{JL}-\mathbf{J}^\intercal(\mathbf{H}-\mathbf{R})\mathbf{1}_d. \tag{33}$$

Equation (30) and (32) give how (17) and (19) are derived. To get (20), we let $(\partial U/\partial\mathbf{R})=0$ and $(\partial U/\partial\mathbf{L})=0$. From $(\partial U/\partial\mathbf{R})=0$, we have

$$\mathbf{GR}+\mathbf{JL1}_d^\intercal=\mathbf{H} \tag{34}$$

$$\mathbf{G}[\mathbf{R}_1,\ldots,\mathbf{R}_d]+\mathbf{J}[\mathbf{L},\ldots,\mathbf{L}]=[\mathbf{H}_1,\ldots,\mathbf{H}_d]$$

$$\begin{bmatrix}\mathbf{G}&&&&\mathbf{J}\\&\mathbf{G}&&&\mathbf{J}\\&&\ddots&&\vdots\\&&&\mathbf{G}&\mathbf{J}\end{bmatrix}\begin{bmatrix}\mathbf{R}_1\\\mathbf{R}_2\\\vdots\\\mathbf{R}_d\\\mathbf{L}\end{bmatrix}=\begin{bmatrix}\mathbf{H}_1\\\mathbf{H}_2\\\vdots\\\mathbf{H}_d\end{bmatrix}. \tag{35}$$

Similarly, from $(\partial U/\partial\mathbf{R})=0$, we have

$$\frac{1}{d}\mathbf{J}^\intercal\mathbf{R}1_d+\mathbf{J}^\intercal\mathbf{JL}=\frac{1}{d}\mathbf{J}^\intercal\mathbf{H} \tag{36}$$

$$\frac{1}{d}\mathbf{J}^\intercal\sum(\mathbf{R}_i)+\mathbf{J}^\intercal\mathbf{JL}=\frac{1}{d}\mathbf{J}^\intercal\sum(\mathbf{H}_i) \tag{37}$$

$$\left[\frac{1}{d}\mathbf{J}^\intercal\ \frac{1}{d}\mathbf{J}^\intercal\ \cdots\ \tfrac{1}{d}\mathbf{J}^\intercal\ \mathbf{J}^\intercal\mathbf{J}\right]\begin{bmatrix}\mathbf{R}_1\\\mathbf{R}_2\\\vdots\\\mathbf{R}_d\\\mathbf{L}\end{bmatrix}=\left[\frac{1}{d}\mathbf{J}^\intercal\mathbf{H}\mathbf{1}_d\right]. \tag{38}$$

Combining (35) and (38), we then get (20).

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